Work Package 9: Unified Monte Carlo and Computational Validation

Reproducibility appendix for causal operator interventions in linear SPDEs

7 July 2026

Abstract

This appendix validates the complete WP3–WP8 theorem chain using an exact spectral simulation of the stochastic heat equation, deterministic Gaussian law calculations, and constructive non-identification experiments. The original Cartesian (N, n) design is audited against the heat-equation resolution condition and replaced, for inferential validation, by a theorem-aligned paired design with 1,000 replications per cell. The suite reports estimator bias and RMSE, mixed-rate LAN diagnostics, counterfactual CDF and quantile errors, pointwise and simultaneous coverage, multiplier sensitivity, Galerkin and temporal slopes, total-error balancing, operator derivatives, and factual-merging/counterfactual-separation failures. A clean one-command execution completed in 67.1 seconds and records raw data, software versions, Git provenance, and SHA-256 checksums.

1 Design audit

For the Dirichlet heat equation on $(0, 1)$,

$$\lambda_N \Delta_n = \pi^2 N^2 / n.$$

Only 1 of 25 planned Cartesian cells satisfy the operational threshold $\lambda_N \Delta_n \leq 0.15$. The high-replication designs are $(5, 2000)$, $(10, 8000)$, and $(20, 32000)$, each with 1,000 replications and $\lambda_N \Delta_n = 0.123370$.

2 Estimation and LAN

N	n	lambda_N_Delta	theta_bias	theta_rmse	theta_empirical_sd	theta_average_se	sigma_bias	sigma_rmse	sigma_empirical_sd	sigma_average_se	ztheta_sd	zsigma_sd	scaled_cross_correlation
5	2000	0.1234	0.003414	0.02416	0.02393	0.0237	6.581e-05	0.002134	0.002134	0.002122	1.018	1.006	0.07463
10	8000	0.1234	0.0004481	0.009086	0.00908	0.008895	2.415e-05	0.0007493	0.0007493	0.0007501	1.022	0.999	0.0966
20	3.2e+04	0.1234	-9.863e-05	0.003253	0.003253	0.003253	6.925e-06	0.0002639	0.000264	0.0002652	0.9995	0.9955	0.1222

The information-standardised empirical standard deviations differ from one by at most 0.0219. Figures 1–3 display the normal and likelihood-ratio diagnostics.

3 Counterfactual targets and bands

N	n	cdf_sup_rmse	effect_sup_rmse	fast_simultaneous_coverage	slow_simultaneous_coverage	fast_average_full_width	slow_average_full_width	pointwise_coverage_min	pointwise_coverage_max
5	2000	0.00172	0.01639	0.946	0.95	0.006707	0.06374	0.945	1
10	8000	0.0006043	0.006286	0.955	0.948	0.002371	0.02417	0.955	1
20	3.2e+04	0.0002129	0.002261	0.951	0.951	0.0008384	0.008856	0.951	1

4 Bootstrap sensitivity

sensitivity_axis	method	draws	grid_size	fast_coverage	slow_coverage	fast_mean_full_width	slow_mean_full_width
multiplier	direct_gaussian	499	31	0.9333	0.93	0.002351	0.02412
multiplier	mode_gaussian	499	31	0.96	0.9467	0.003021	0.03424
multiplier	mode_rademacher	499	31	0.9567	0.9467	0.002943	0.03279
draw_count	direct_gaussian	199	31	0.94	0.9367	0.00235	0.02385
draw_count	direct_gaussian	999	31	0.94	0.9267	0.002375	0.02415
grid_resolution	direct_gaussian	499	11	0.9333	0.9267	0.002361	0.02407
grid_resolution	direct_gaussian	499	61	0.9367	0.92	0.002369	0.0241

The mode-block procedures are mildly conservative at finite N, while the direct Gaussian multiplier is narrower because the limiting covariance is known analytically.

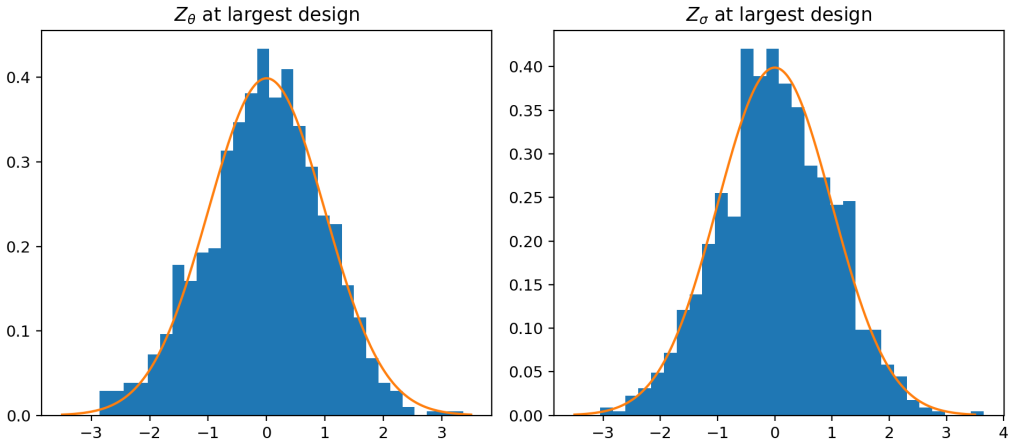


Figure 1: Information-standardised estimator histograms at the largest theorem-aligned design.

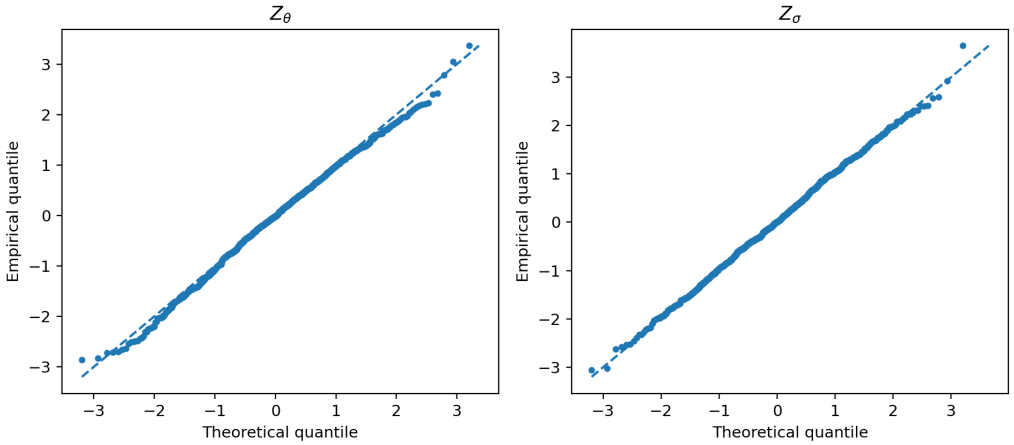


Figure 2: Normal quantile diagnostics for the mixed-rate estimators.

5 Galerkin, temporal, and total error

alpha_q	metric	predicted_slope	empirical_slope	ci_low	ci_high
0	mean_error	-2.6	-2.58	-2.595	-2.565
0	variance_error	-4.6	-4.564	-4.595	-4.533
0	strong_rms	-2.3	-2.307	-2.311	-2.302
0	cdf_sup	-2.6	-2.58	-2.595	-2.565
0.75	mean_error	-2.6	-2.58	-2.595	-2.565
0.75	variance_error	-6.1	-6.082	-6.259	-5.905
0.75	strong_rms	-2.6	-2.587	-2.598	-2.575
0.75	cdf_sup	-2.6	-2.58	-2.595	-2.565
1.5	mean_error	-2.6	-2.58	-2.595	-2.565
1.5	variance_error	-7.6	-8.282	-15.38	-1.185
1.5	strong_rms	-2.6	-2.58	-2.595	-2.565
1.5	cdf_sup	-2.6	-2.58	-2.595	-2.565

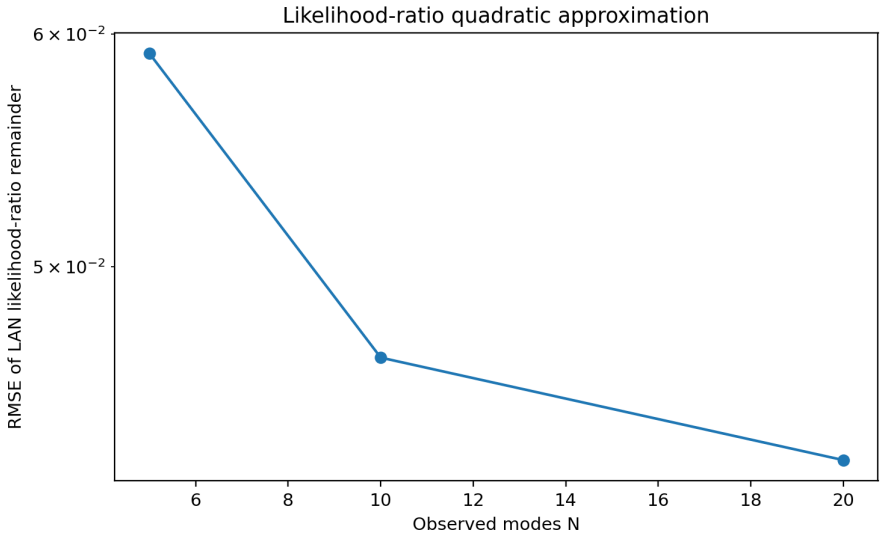


Figure 3: RMSE of the LAN likelihood-ratio quadratic remainder.

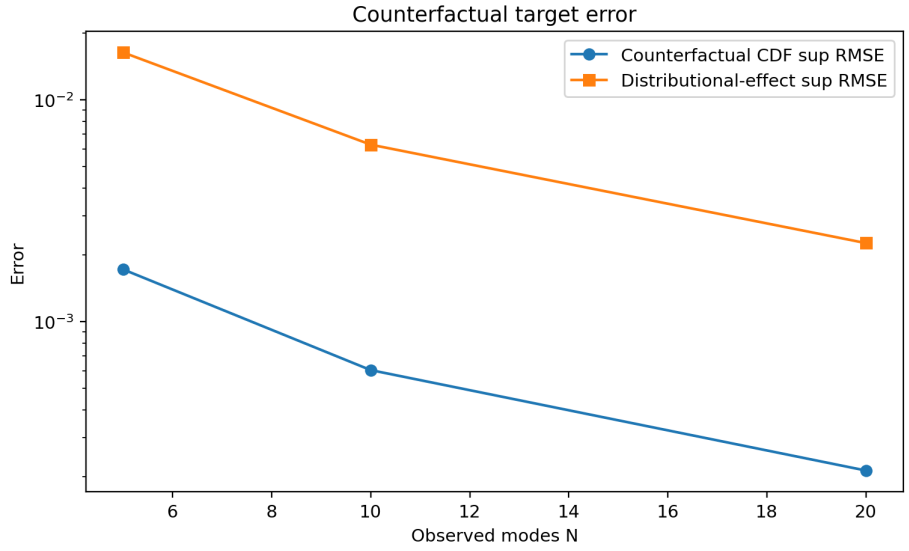


Figure 4: Contraction of fixed-intervention CDF and factual-versus-counterfactual effect errors.

6 Operator sensitivity

The central difference error falls from 4.270e-06 to 1.708e-09. The first-order CDF remainder is quadratic and the Wasserstein distance stays below the common-noise bound.

7 Non-identification

The finite-grid open-loop example has factual covariance discrepancy 8.185e-16 and post-intervention discrepancy 0.7003.

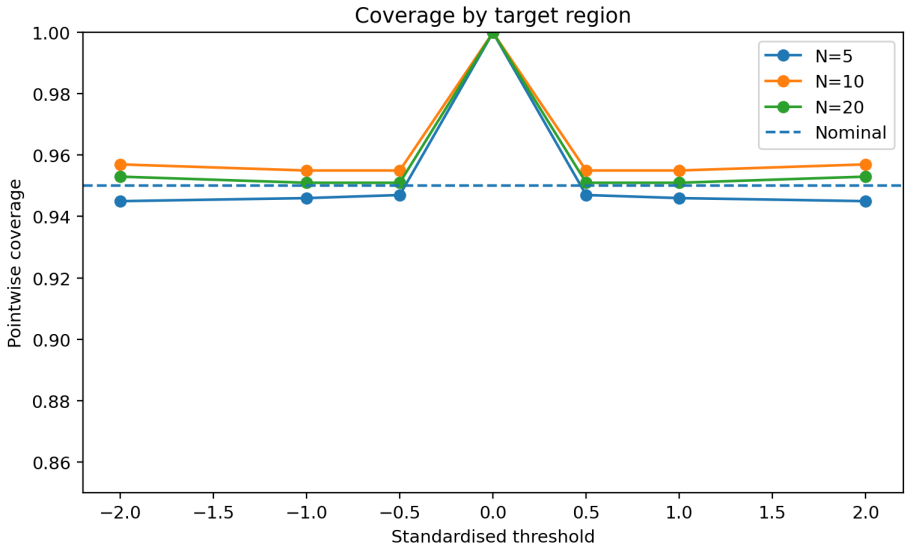


Figure 5: Pointwise coverage over central and tail threshold regions.

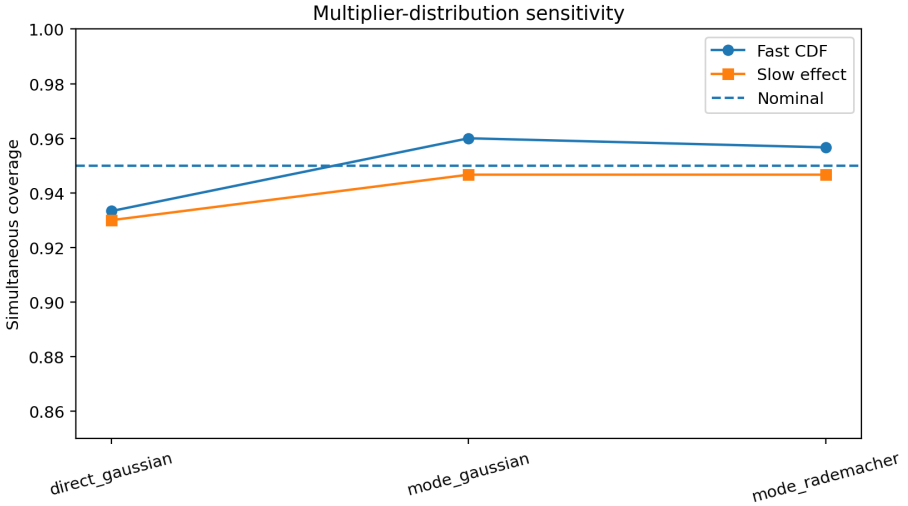


Figure 6: Simultaneous coverage by multiplier implementation.

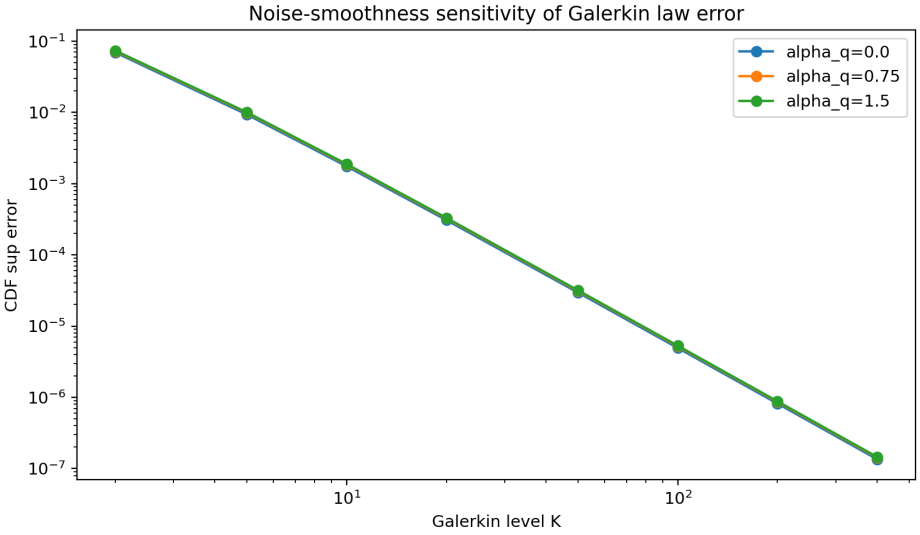


Figure 7: Noise-smoothness sensitivity of the Galerkin CDF-law error.

8 Completion gate and reproducibility

criterion	passed	diagnostic
theorem_aligned_designs	True	0.1234
at_least_1000_reps	True	1000
standardised_sd_near_one	True	0.02187
simultaneous_coverage_reported	True	7
rate_slopes_checked	True	0.6822
operator_remainder_checked	True	4.675e-08
failure_cases_demonstrated	True	0.7003
planned_grid_audited	True	1
one_command_runner	True	1

The suite is executed with `make wp9`. It stores the full configuration, raw estimator replicates, summary tables, figures, environment metadata, the source commit, and SHA-256 checksums.

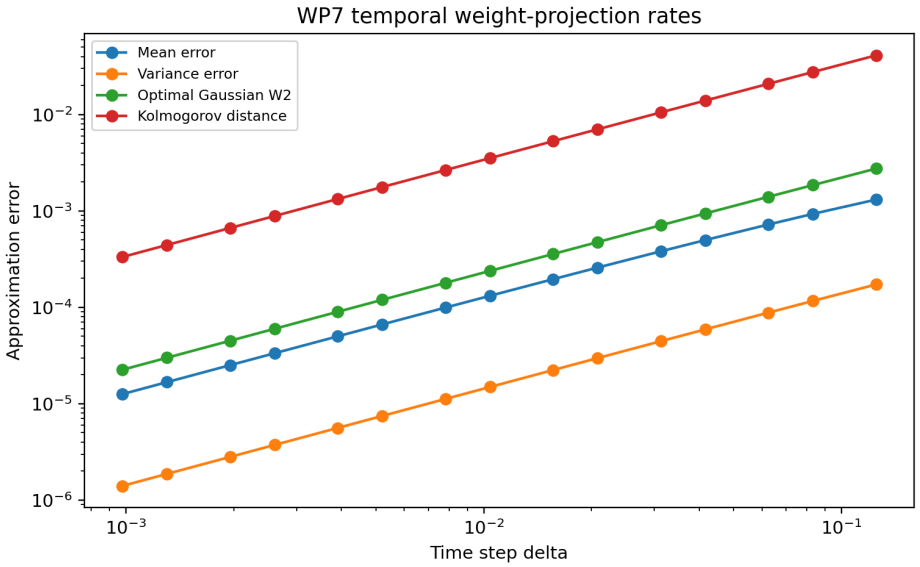


Figure 8: First-order temporal approximation rates.

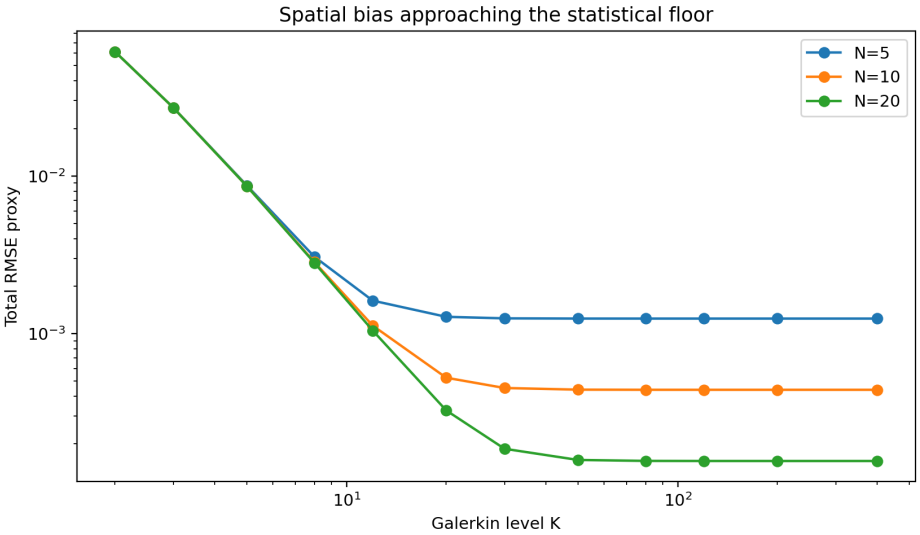


Figure 9: Truncation bias approaching the statistical error floor.

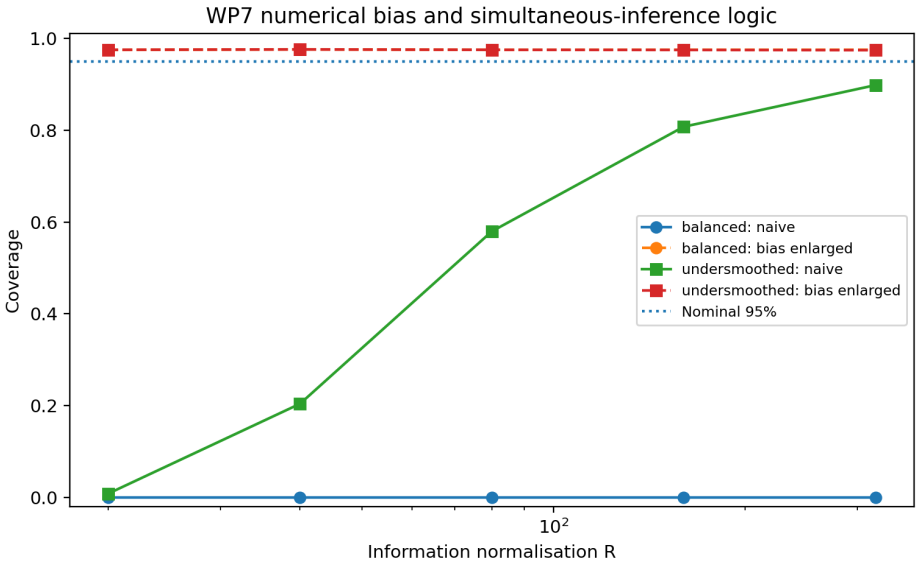


Figure 10: Naive and certified-bias-enlarged coverage.

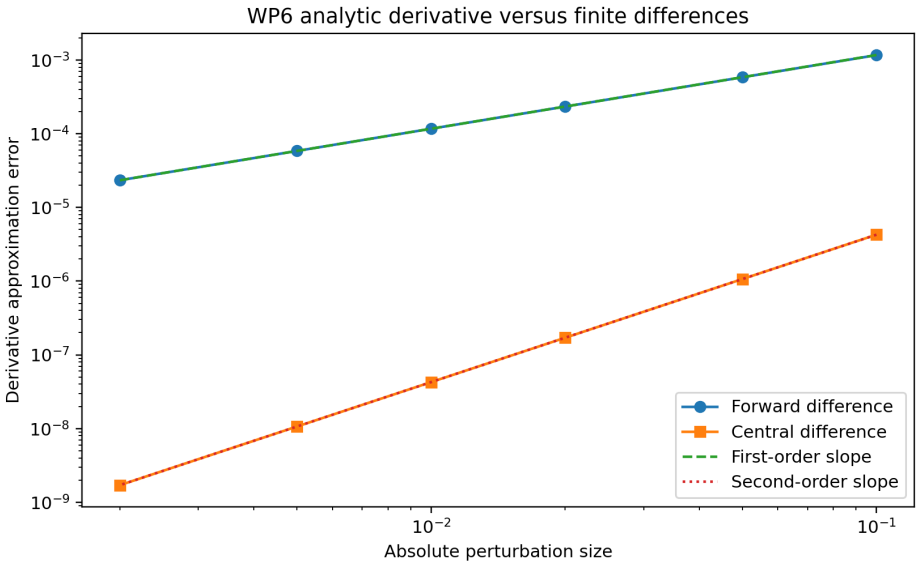


Figure 11: Finite-difference validation of the operator-law derivative.

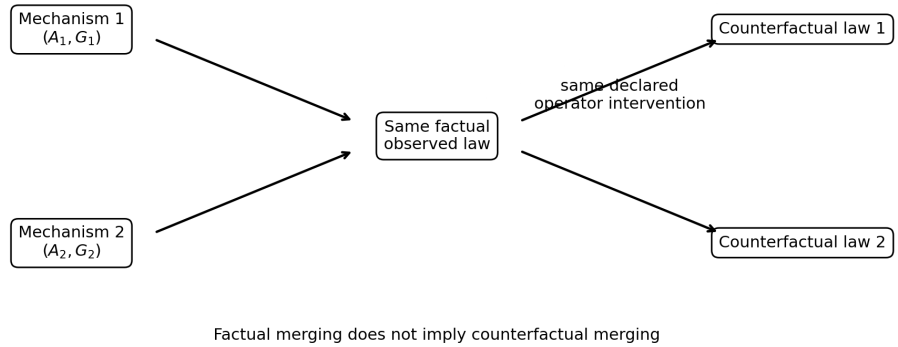


Figure 12: Factual merging followed by counterfactual separation.

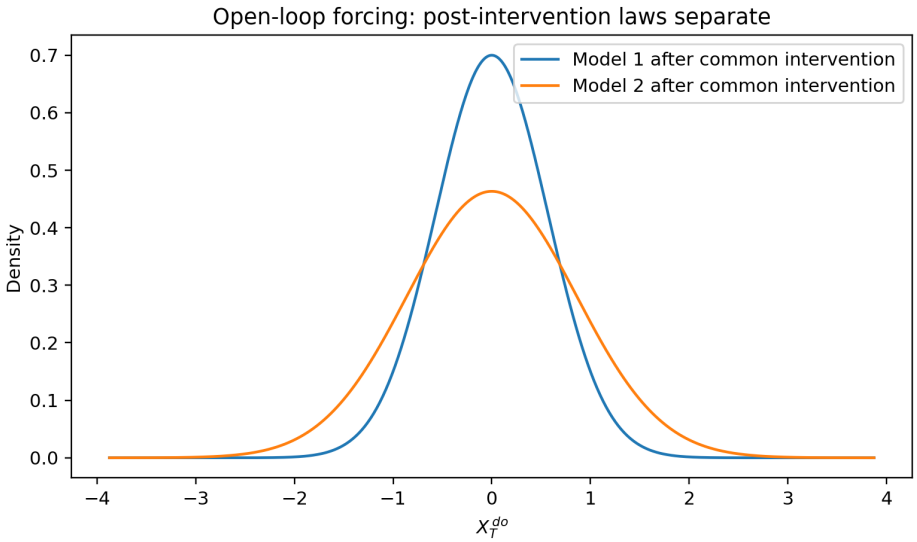


Figure 13: Open-loop forcing: separated post-intervention laws after exact factual equivalence.